Library Assistant

AI-Powered Support for AUB Libraries

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Introduction

This capstone project builds an AI-powered assistant for the AUB Libraries using Retrieval-Augmented Generation (RAG). It provides bilingual support in Arabic and English, answers library FAQs, recommends academic databases, and surfaces information from the library website — all grounded in verified sources.

Bilingual

Arabic & English with per-message language detection and RTL support

RAG-Powered

Hybrid retrieval combining semantic vector search with keyword matching

Admin Dashboard

Librarians manage data, review analytics, and correct answers

24/7 Availability

Always-on support reducing repetitive reference desk inquiries

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Information about Client

1866
Year Founded

1,000+
Electronic Databases

6+
Library Branches

450+
Subject Databases

- AUB is a leading research university in the Middle East, serving thousands of students, faculty, and researchers
- Libraries span multiple branches: Jafet, Science & Engineering, Saab Medical, and more
- Services include book loans, room reservations, interlibrary loans, printing/scanning, research consultations, and database access
- Bilingual community where students interact in both Arabic and English daily
- Librarians handle high volumes of repetitive questions at reference desks and via email

Problem Statement

Repetitive Workload

Staff spend significant time answering the same FAQs — borrowing policies, hours, database access, printing

Scattered Information

Answers live across FAQ pages, database catalogs, and dozens of web pages — students struggle to find what they need

No 24/7 Support

Reference desk operates during business hours only; students studying late or remotely have no way to get answers

Language Barrier

Arabic-speaking students may not find answers on the English-dominant library website

Database Discovery Gap

With 450+ databases, students often don't know which are relevant to their research area

No Feedback Loop

Librarians have no systematic way to see what students ask or what questions go unanswered

Literature Review / Benchmarking

- **RAG:** Lewis et al. (2020) introduced RAG, showing that grounding LLM responses in retrieved documents dramatically reduces hallucination
- **University chatbots:** Several universities deployed rule-based chatbots (e.g., Jill Watson at Georgia Tech), but these require manual intent mapping
- **Vector search:** pgvector and similar tools enable semantic search over domain-specific corpora, outperforming keyword-only approaches (BM25)
- **Hybrid retrieval:** Combining dense (vector) and sparse (keyword) retrieval via Reciprocal Rank Fusion yields better recall (Ma et al., 2023)
- **Multilingual RAG:** Cross-lingual retrieval (Arabic query → English KB) loses 10-20% similarity signal; query rewriting bridges languages

Approach	Semantic Understanding	Grounded	Bilingual
Basic FAQ Bots	No	Yes	No
Generic LLMs (ChatGPT)	Yes	No	Partial
Our System	**Yes**	**Yes**	**Yes**

Data Sources and Details

Source	Records	Description
Library FAQ CSV	~184 Q&A pairs	Curated FAQ covering borrowing, reservations, fines, printing, WiFi, ILL, room booking, etc.
Database Descriptions CSV	~455 databases	Name + description for each electronic database (ABI/INFORM, PubMed, IEEE, etc.)
Scraped Library Web Pages	Dozens of pages	Playwright-scraped AUB library website — hours, policies, services, contact info
Admin Feedback Corrections	Ongoing	Librarian-corrected answers stored with embeddings for future retrieval
Chat Conversation Logs	Ongoing	Every query, answer, language, scores, and retrieved chunks — stored in PostgreSQL

Data Processing Pipeline

Playwright Scraper → HTML Cleaning → Structure-Aware Chunking → OpenAI Embeddings → PostgreSQL + pgvector

Solution

An AI-Powered Library Assistant that:

- J fj
    - Uses Retrieval-Augmented Generation (RAG)
    - Retrieves relevant library data before generating answers
    - Understands natural language queries (including follow-ups)
    - Works bilingually (Arabic & English)
    - Improves over time using feedback from librarians

How it works:

1. User submits a question
2. Query is processed and refined
3. Relevant data is retrieved from multiple sources
4. Results are ranked and filtered
5. Answer is generated using retrieved context
6. Output is verified before being shown

Solution

System Components:

1. Student Interface
    1. Chat-based UI
    2. Arabic + English support
    3. Displays structured answers
2. Admin Dashboard
    1. Manage data (FAQs, databases)
    2. Monitor system performance
    3. Review conversations and feedback
3. Ingestion Pipeline
    1. Web scraping system
    2. Data processing and embedding
    3. Automatic indexing

Key Technical Decisions:

1. Hybrid retrieval (vector + keyword search)
2. Query rewriting for follow-ups and Arabic queries
3. LLM-based reranking for better answer selection
4. Post-generation verification to reduce incorrect outputs
5. Source prioritization to prefer reliable data

RAG Pipeline Architecture

1. Input Guards: Regex + embedding-based prompt injection detection (two-tier) + domain scope filtering (blocks off-topic/ malicious queries before any API call)
2. Query Rewriting: LLM-based (GPT-4o-mini): resolves follow-ups using conversation history, translates Arabic→English for retrieval
3. Cache Lookup: Exact match on (rewritten query, language) returned immediately. On miss, semantic similarity check (cosine ≥ 0.95) against stored query embeddings — near-duplicate queries served from cache without any retrieval or LLM calls.
4. Admin Feedback Correction: pgvector similarity search (≥ 0.85) against librarian-corrected answers. If a match is found, the corrected answer is used directly — bypassing the full retrieval pipeline.
4. Intent Classification: Keyword/regex-based routing: determines which tables to search (FAQ, databases, library pages) and applies page_type filters
5. Hybrid Retrieval: Per table: vector search (cosine via pgvector) + keyword search (phrase + tsquery with synonym expansion). Merged via RRF (65%/35%)
6. LLM Reranking: Deduplicate (Jaccard 0.85), top 15 candidates, score relevance 0–1 with GPT-4o-mini, filter below 0.5
7. Grounded Generation: Role-locked as AUB librarian, must cite or remove claims, temperature=0.0. Context sufficiency check with confidence thresholds
8. Claim Verification: Post-generation LLM check + regex safety validation. If verifier errors out, returns cautious fallback — never raw unverified text

Tools Developed

Chat Interface (Student-Facing)

- Bilingual React interface (Arabic RTL support)
- Markdown rendering for formatted answers
- Debug panel: retrieval scores, sources, pipeline stages
- Real-time language detection
- Human escalation: students can submit unanswered questions to a librarian

Admin Dashboard (Librarian-Facing)

- System status & health monitoring
- Full CRUD for FAQs and databases, one-click reindex
- 22 auto-generated analytics charts
- Admin feedback/correction system
- Escalations tab: librarians review and respond to student escalations

Ingestion Pipeline

- Playwright web scraper for AUB library site
- Structure-aware chunking (tables intact, FAQ pairs as chunks)
- Automated embedding generation & pgvector indexing

Analysis Results & Key Findings

- **Hybrid retrieval outperforms pure vector search:** Adding keyword search with synonym expansion improved recall for specific library terms
- **Query rewriting resolves 30%+ of queries:** Follow-up questions, Arabic queries, and vague inputs are rewritten into effective retrieval queries
- **Cross-lingual threshold offset is critical:** Arabic queries score 10-20% lower in cosine similarity; without the 0.10 offset, valid Arabic questions are incorrectly rejected
- **Claim verification catches hallucinations:** Post-generation verifier removes unsupported claims; fail-safe ensures errors never surface unverified text
- **Intent-based pre-filtering reduces noise:** Routing queries to relevant page types dramatically improves top-result relevance
- **Caching reduces latency and API cost:** Repeated/similar queries hit the in-memory cache, avoiding redundant embedding + LLM calls
- **Admin feedback loop closes the gap:** Librarian corrections are retrieved directly for future similar queries (similarity ≥0.85)

Limitations:

1. Performance depends on data quality
2. Arabic retrieval still slightly weaker than English
3. Confidence thresholds are manually tuned
4. No large-scale evaluation dataset yet

Future Work:

1. Build evaluation dataset to measure accuracy
2. Improve multilingual retrieval
3. Expand data coverage
4. Deploy system for real users

Thank You

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